

# Mohammed Azharudeen Farook Deen

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📁 Portfolio Website    in fmdazhar    🌐 fmdazhar

## About

Robotics Engineer with a focus on **deep robot learning** and **optimal control** to build intelligent and safe robots.

## Skills

**Programming:** Python, C++, MATLAB

**Machine Learning & AI:** PyTorch, JAX, TensorFlow, Deep Learning (CNNs, RNNs, Transformers), Reinforcement Learning (PPO, SAC), Imitation Learning (BC, HG-Dagger), Robot Foundation Models (NVIDIA GROOT, pi-zero)

**Robotics & Simulation:** ROS, ROS2, Isaac Sim, Isaac Gym, Mujoco, NVIDIA Warp

**Optimization & Control:** Model Predictive Control (MPC), Quadratic/Nonlinear Programming (QP, SQP)

**Development Tools & Systems:** Git, Docker, Linux

**Languages:** English (Business Fluency), German (Beginner), Tamil (Native)

## Work Experience

### Intern

Fraunhofer IPA

Stuttgart, Germany

July 2024 – Mar 2025

- Developed a **digital twin simulation environment in MuJoCo and ROS2**, supporting both state- and vision-based inputs for a peg-in-hole assembly task with different difficulty presets.
- Implemented and tuned various controller modules; Integrated teleoperation modules for keyboard and spacemouse control for **Learning from Demonstrations (LfD)** and **Human-in-the-Loop RL**.
- Built a real robot interface with **Flask-ROS server** for command execution with **actor-learner asynchronous nodes**.
- Adapted both the simulation and real environment to work with real-time online Reinforcement Learning (SAC) and Imitation Learning (BC, HG-Dagger) pipelines in both **Pytorch** and **JAX** and benchmarked them using simulation.

### Master Thesis - Robot Learning of Quadrupedal Locomotion on Deformable Terrain

RWTH Institute for Data Science in Mechanical Engineering (DSME) and RWTH Institute of Geomechanics and Underground Technology (GUT)

Aachen, Germany

Sep 2023 – Mar 2024

- Integrated a particle-based contact model to simulate the complex behaviors of granular materials in **NVIDIA Isaac Sim**, developed an adaptable and robust controller using Reinforcement Learning (PPO) in PyTorch for quadruped (Unitree A1) to follow a unified policy in diverse natural environments and evaluated the performance with sim-to-real additions.
- Preliminary work accepted at the 1st German Robotics Conference (GRC2025), and a related paper is being finalized for publication.

### Student Assistant

RWTH Institute for Automatic Control (RWTH IRT)

Aachen, Germany

May 2023 – June 2024

- Extended codebase of a unified C++ interface for various **quadratic programming solvers (QP)** to **nonlinear programming (NLP) solvers** and included coupled tools for MPC, NMPC, SQP, auto differentiation, Hessian regularization methods, etc with appropriate test cases.
- Involved in the Fernbin project, where my tasks included developing simple simulation for testing control systems in Python and ROS2 for ships that react dynamically to the surroundings.

### Research Project

Automated and Connected Driving Challenges — RWTH Institute for Automotive Engineering (ika)

Aachen, Germany

Apr 2023 – Aug 2023

- Enhanced trajectory planning in autonomous vehicles using **Model Predictive Control (MPC)**.

### Teaching Assistant

RWTH Institute of Geomechanics and Underground Technology (RWTH GUT)

Aachen, Germany

Apr 2023 – Sep 2023

- Implemented engaging programming exercises, conducted tutorial sessions, fostering student understanding of key robotics programming concepts in Python and ROS for the course “Introduction to Robotics”.

### Research Assistant

Microscale Transport Laboratory, Indian Institute of Information Technology (IIITDM))

Kancheepuram, India

Aug 2019 – Mar 2021

- Studied predominant microscale phenomena subjected to acoustic fields and published results in various reputed journals.

## Education

**RWTH Aachen University**  
*Master of Science in **Robotic Systems Engineering***

*Oct 2021 – Dec 2024*

**Shiv Nadar University**  
*Bachelor of Technology with major in **Mechanical Engineering**(Specilization: Computational Techniques) and minor in Electronics and Communication Engineering*

*Aug 2015 - May 2019*

## Journal Publications and Accepted Conferences

**Rapid Quadrupedal Locomotion on Deformable Terrain - 1st German Robotics Conference**

March 2025

*Mohammed Azharudeen Farook Deen, Omer Kemal Adak, Raul Fuentes*  
<https://www.robotics-institute-germany.de/conference/> 

**Theory of nonlinear acoustic forces acting on inhomogeneous fluids. Journal of Fluid Mechanics. [IF(2022)-3.7]**

April 2022

*Rajendran VK, Jayakumar S, Azharudeen M, Subramani K*  
<https://doi.org/10.1017/jfm.2022.257> 

**Heat transfer mechanism driven by acoustic body force under acoustic fields. Physical Review Fluids. [IF(2021)-2.5]**

July, 2021

*Azharudeen M, Kumar V, Pothuri C, Subramani K*

**A Novel Heat Transfer Mechanism Using Acoustic Waves - Presented at International Heat and Mass Transfer Conference, IIT Roorke**

Dec 2019

*Azharudeen M, Pothuri C, Subramani K*

<https://doi.org/10.1103/PhysRevFluids.6.073501> 

**Rapid mixing in microchannel using standing bulk acoustic waves. Physics of Fluids. [IF(2020)-3.5]**

Dec 2019

*Azharudeen M, Pothuri C, Subramani K*

<https://doi.org/10.1017/jfm.2022.257> 

## Honors & Awards

**Full Academic Scholarship, 8 Semesters – Shiv Nadar University**

*Aug 2015*

*Issued by Shiv Nadar University*

Completed my engineering program with 100% scholarship across all 8 semesters.